

Acceptance requires exactly one compatible candidate.

OUTCOME	ROWS	DISTINCT IDS
Matched	61	60
All candidates contradicted	40	40
Ambiguous	10	10
Conflicting source data	1	1
No candidate in reference	0	0

112 worksheet rows · 111 distinct vehicle IDs.
Unresolved vehicles stay in the source denominator and never enter the VIO sheet.

DECISION POLICY IN THIS RUN

Version criteria are capacity, fuel and power; variant criteria are body, engine and drive. Missing evidence is unknown and never becomes agreement.

A documented manufacturer model code decoded from a scoped identifier can make a sole compatible candidate sufficient; a documented disagreement is a veto. Extra agreements never eliminate a compatible sibling.

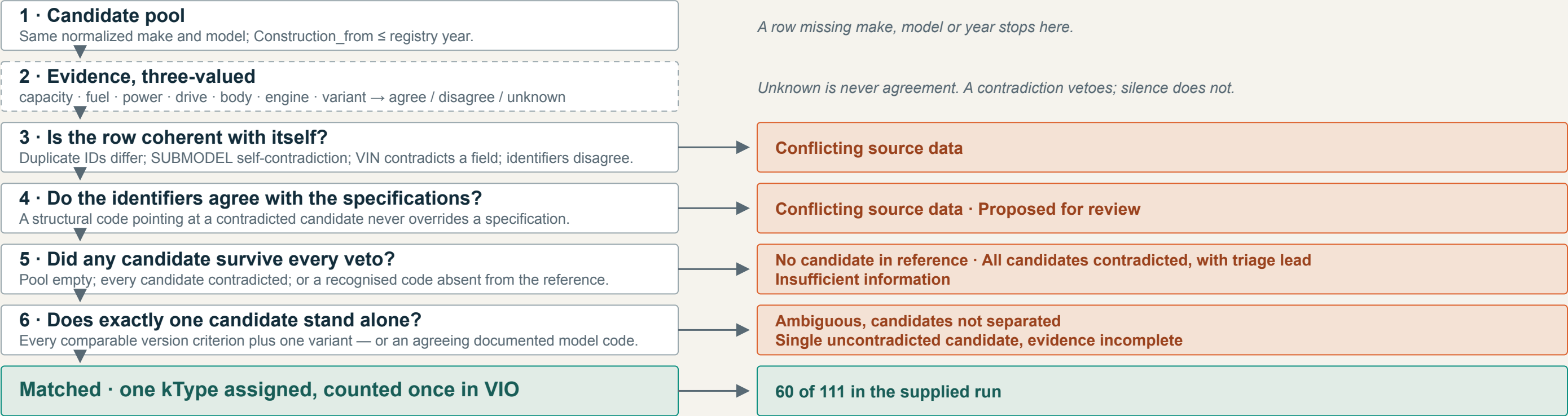
Year mode is **gate**: a candidate whose `Construction_from` year is later than the registry year is excluded.
`Construction_to` does not gate, because the NZ year may be first registration.

Exact power agreement · 0 tolerance used

The one-kilowatt band orders review only.
No acceptance in this run depends on it.

A row stops at the first thing wrong with it.

Seven stages, in this order. Six of them can only stop a row; one can assign a kType.



The order is load-bearing: every stage returns, so a row is reported against the first obstacle it meets rather than the last. The default policy reads no weights, no threshold and no score.

Each unresolved group names a field and both values.

GROUP	FIELD	SOURCE VS REFERENCE	OUTCOME	IDS
X-TRAIL ST 2.5P/6CVT	power	125 kW vs 126 kW (kTypes 107383, 107384); 27 of 28 candidates	All candidates contradicted	20
RANGER WILDTRAK 2.0D/4WD	capacity; power	1996 cc vs 2184–3198 cc; 157 kW vs 57–147 kW across 13 candidates	All candidates contradicted	14
C-CLASS C 200 1.5P/9AT	capacity	Power, fuel, variant and engine agree with kType 137964; capacity alone disagrees, 1491 cc against 1991 cc	All candidates contradicted	6
COLORADO LS RC SC 2.8D/4WD	—	kTypes 100868 and 100871 both remain compatible	Ambiguous	10
C-CLASS C 200 1.5P/9AT	capacity	Same, plus the VIN <i>Grundbaumuster</i> 205380 authoritatively confirms 137964 against the source's own capacity	Conflicting source data	1

Three different problems: a publisher disagreement, an absent catalogue version, a registry recording error. Naming which is which is what a reviewer acts on.

The rule I deleted, and why.

Deleted in v2 with an unsourced VIN comparison: eighteen assignments lost, recorded in the shipped rule table.

Removed from the body table

NZ STATION WAGON → Estate / SUV / Closed Off-Road Vehicle

NZ SPORTS CAR → Coupe / Convertible / Saloon

NZ UTILITY → Pickup / Platform/Chassis

Judgement, not vocabulary

No publisher states these correspondences. The two taxonomies are independent, so the table was asserting my opinion with the authority of a reference.

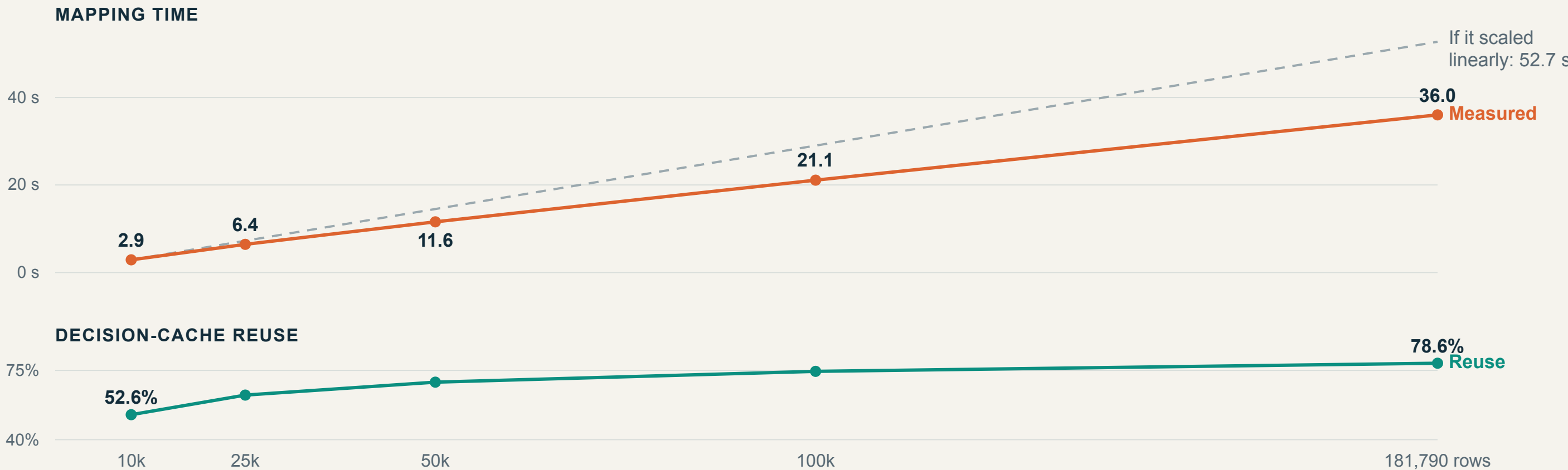
What survives is name identity only. A registration body pairs with a reference structure **solely where both publishers print the same word**: NZTA VIRM Table 2-2-8 names SL "Saloon" and CV "Convertible"; TecDoc KT 086 states "Saloon" and "Convertible" in the supplied extract. A reviewer can check both sides. Nothing else is paired.

Covers 2 of NZTA’s 26 body types. The rule file declares that incompleteness itself, so a gap is never read as evidence: in this run the body field agrees or stays silent, and has never once contradicted a candidate.

The cost is visible in the coverage figure and nowhere hidden: removing these and an unsourced VIN comparison cost eighteen C-Class assignments. A sourced chassis rule in the same commit added nine, so the net was 51 to 42, before sourced identifier evidence recovered past it. That is the trade this project makes everywhere.

Cost grows more slowly than volume.

The published 2020 New Zealand register, not supplied with the task: 36 seconds to map, under a minute to a complete result set as CSV. Eighteen times the rows cost twelve times the time.



Source rows, linear scale. Median of three runs per point; run-to-run spread at most 4.6%. The published file is unordered, so each prefix is representative.

The matcher is not the bottleneck. Workbook serialization is.

PHASE · 181,790 ROWS · ROUNDED	CSV OUT	WITH XLSX
Load and normalize source	1.6 s	1.5 s
Map and stream evidence	36.2 s	35.9 s
Build sheets	7.5 s	7.5 s
Write sheets as CSV	10.1 s	9.5 s
Write workbook (XLSX)	—	430.0 s
Run metadata (hashing)	0.2 s	0.2 s
Write performance report	3.7 s	3.5 s
Total	59.3 s	488.0 s

WHAT THIS SAYS TO DO

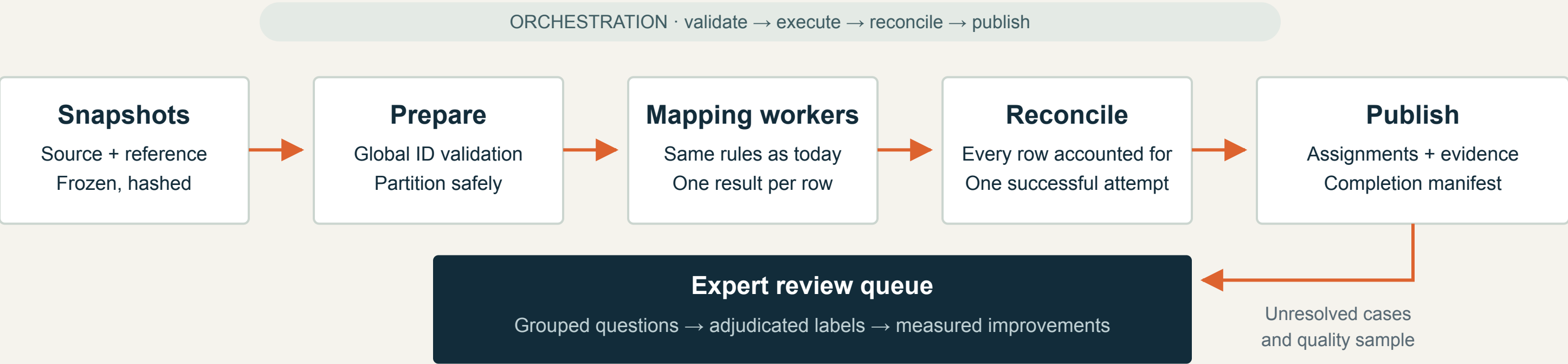
XLSX serialization is 88% of wall clock and 45× the cost of the identical data as CSV, because Results and Review_Detail carry ~360,000 wide rows.

The fix is the container, not the code: route row-level sheets to CSV above a threshold, exactly as candidate evidence already is, and keep the workbook for the sheets a person reads. That returns a full run to about a minute.

Not done, deliberately. At the 111-row scale of this submission the workbook takes about five seconds and is the right format.

Scale the work. Keep the decisions traceable.

A sketch for the costed expansion plan, if and only if the pilot passes all three gates.



Retries: separate attempt outputs **Publication:** manifest written after reconciliation **Adjudication:** the only source of quality labels

Adding workers does not resolve oversized candidate groups. Those need a validated additional retrieval or comparison pass.

Reviewer corrections become rules.

That is where the learning belongs. Each stage produces what the next one needs, and a model never assigns a kType.

STAGE 0 · THE PILOT

Adjudicate

Expert decisions on a real batch. This is the measurement, and it produces the first labelled data this problem has ever had.

Gate: precision meets the agreed floor and effort falls.

STAGE 1 · CLASSICAL ML

Induce rules

Mine adjudications for readable conditions (*where BODY_TYPE is UTILITY and the cab code is X, reviewers chose Pickup*), each carrying its support count and confidence. Proposed to the owner, not applied.

Plus a learned ranker for review order. No language model.

STAGE 2 · BEDROCK

Explain exceptions

Natural-language explanation of one conflict, from evidence already computed. Measured on reviewer effort, never on accuracy.

Stage 3, a learned scorer, only if 1 and 2 pay.

Per row and high volume → rules or a small classical model. Per exception and read by a human → a language model. A per-row LLM call across 181,790 vehicles is indefensible; across 12,677 exceptions it is bounded.

Sixty accepted rows are four rules.

The cache key splits on fields the outcome never reads. Count the sample in what can independently be wrong: the kType and the basis it was accepted on.

RULES ADJUDICATED, NO ERRORS	RULES OUT PRECISION WORSE THAN
4 · the whole accepted set today	0.47
30	90%
59	95%
99	97%
149	98% · an output, never the gate

One-sided 95% Clopper–Pearson lower bound, zero errors found. Nine cache keys stand behind these four rules, so every figure is a ceiling.

TWO GATES, TWO SAMPLES

Precision 0.90 on accepted rules — at 150 it tolerates 8 genuine errors. **Recall 0.85** on the unresolved queue, deliberately looser: a missed match is visible in the review queue, a wrong one is not.

SIDE	ROWS	DISTINCT DECISIONS
Accepted · this extract	60	9 keys · 4 rules
Unresolved · this extract	51	9 keys · 5 causes
Unresolved · full register	12,677	412

Count the batch first. 835 vehicles share one decision; 237 of 483 appear once.

Three open questions, re-tested one at a time.

Each row was measured by re-running the mapper with one change applied in a scratch copy. No shipped rule, profile or data file was modified.

CHANGE APPLIED	ACCEPTED	WHAT IT STILL NEEDS
Baseline · as submitted	60	—
Ruling 1 · 125 kW counts as 126 kW	80	A data-owner ruling — the tolerance alone moves statuses, not assignments
Colorado · either half alone	60 · unchanged	Catalogue fix alone does nothing; VIN profile alone degrades the ten further
Colorado · both together	70	They have to land in that order
Mercedes · capacity 1491 → 1991	61, not 67	Six rows then face four identical C 200 bodies
Mercedes · capacity plus one identifier format	67	Mercedes' W1K layout, or NZTA documenting MVMA
Ranger · add the four real kTypes	60 · Ambiguous	Three candidates agree on every comparable field

The first three together reach **97 of 111** — the brief’s ceiling. The last fourteen do not close: a sourced Ford VIN layout is the rest of the answer.

Same engine. 48× the reference. 2.2× the mapping time.

The same 181,790 NZ registrations and the same decision code, tested against a separately obtained reference containing 17,710 distinct kTypes.

FULL 2020 REGISTER	SUPPLIED RDM	LARGER REFERENCE
Reference size	365 rows	17,710 kTypes
No candidate in reference	166,519	62,508
Accepted · shipped rules	2,593	25,420
Accepted · rulings simulated	5,282	28,124
Ambiguous · rulings simulated	—	40,107
Mapping time	36.0 s	79.4 s

WHAT CHANGED

Ambiguity became the constraint

With the open rulings simulated, 40,107 vehicles had candidates the registry evidence could not separate — 1.4× the number accepted in that same run.

The next marginal gain comes from sourced VIN layouts and chassis codes, not from more compute or more catalogue rows.

On the supplied batch, both references accepted the same 60 vehicles with no assignment disagreement. That is a consistency check, not an accuracy estimate.

Regional VIO needs a floor per cell, not per country.

TecAlliance delivers VIO down to NUTS regional levels. This output is national: kType and a count. The registry's regional field survives untouched, so the dimension is a grouping — but the statistics do not survive the split.

CHECKED AGAINST THE SUBMITTED WORKBOOK

VIO sheet	kType × count · no region
TLA carried into Results	yes, verbatim · 34 distinct
Accepted rows carrying kType <i>and</i> region	60 of 60
region as a declared role	not one of the 15
kType × region cells from 60 vehicles	31
Cells holding exactly one vehicle	24 of 31

The regional field rides through because Results reproduces every supplied column unchanged. It is a passenger, not something the matcher understands.

WHY THE FLOOR HAS TO BE RE-AGREED

Every precision figure in this pack is computed on the **national** aggregate. Split to NUTS-3 and there is no averaging left: one wrong kType moves a whole cell. A national floor of 90% says nothing useful about a cell of one.

New Zealand has no NUTS codes. Mapping a Territorial Authority onto a NUTS region would be a correspondence no publisher states — the class of rule deleted in A4, and the same argument applies.

For an EU register the problem mostly disappears: the register carries the regional code, and it becomes a declared role plus a second grouping key.

Remove duplicate work before adding compute.

The all-years NZ fleet moved the bottleneck from file output to candidate evaluation. The decision policy does not need to change.

OBSERVED · BROADER REFERENCE

1,907 rows/s

5.9m registry rows 51.6 minutes

1.83m unique decisions 69% reusable rows

102.6m logical candidate comparisons. No XLSX was written.

01

Columnar intake

Project required fields into normalized Parquet; encode repeated text once.

02

Bulk candidate gate

Vectorized make/model/year join; fast-path the 2.52m rows with no candidate.

03

Decide once

Map each distinct configuration once, keeping a compact outcome cache.

04

Partition & join

Parallelize by make/model, then join multiplicities back for VIO and audit.

Acceptance gate: ≥5,000 rows/s · full run <20 min · peak memory <4 GB · identical kType and status counts.

Sequence matters: implement grouping, bulk gating and compact keys first; add parallel workers only after the single-process path stops repeating work.